

HF SIGNAL 01: A Candidate Machine-Readable Protective Signal for Testing and Standardisation

Technical Parameters, Legal Limits and Validation Requirements

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Abstract

The prohibition on attacking a person recognised as hors de combat forms part of customary international humanitarian law (Rule 47 of the ICRC Customary IHL Study; Article 23(c) of the 1907 Hague Regulations) and binds all parties to an armed conflict irrespective of treaty ratification. For States party to Additional Protocol I, Article 41 provides the clearest treaty formulation of that protection, while Article 57 requires those planning or deciding upon an attack to take every feasible step to verify that the attack remains lawful. Those obligations are long established. Autonomous and semi-autonomous systems operating at machine speed nevertheless create a practical difficulty because there is no commonly specified machine-readable form through which an intention to surrender can be communicated to, and recognised by, such systems.

This paper presents HF SIGNAL 01, a candidate protocol for surrender recognition intended to serve as a technical referent for machine-readable expressions of surrender. It offers two alternative modalities. One is a passive visual modality requiring no equipment. The other is an active electronic modality made up of two channels that must be confirmed together — a pulsating white optical signal and a radio-frequency signature. The paper makes no claim of operational effectiveness, adoption or legal status. The protocol is a supplementary means of expressing an intention to surrender, never a condition of protection, and its absence supports no adverse inference. The proposal is instead assessed against feasibility criteria previously defined for lawful operational safeguards, including technology readiness, latency, annotated datasets and independent validation.

The assessment is intentionally conservative. HF SIGNAL 01 currently exists as a publicly documented candidate whose active modality is specified closely enough to permit experimental characterisation, but without experimental validation, implementation or performance data. Its contribution at present is documentary rather than operational. The paper argues that the existence of a publicly available, integrity-protected candidate standard changes the factual setting in which surrender-recognition capability may be assessed. What was previously described solely as a technical impossibility becomes a documented question of design choice, review and validation. It concludes by inviting independent technical testing, dataset development and examination in the relevant fora of international humanitarian law and autonomous-weapons governance.

Keywords: international humanitarian law; autonomous weapons systems; surrender recognition; hors de combat; Article 41 Additional Protocol I; lawful operational safeguards; machine-readable signals; CCW GGE.

1. The Legal Framework and the Effectiveness Gap

An earlier paper by this author defined a lawful operational safeguard as a technical measure that implements, within the operational architecture of an AI system, an obligation already binding under international humanitarian law — as distinct from safeguards reflecting discretionary ethical preferences, reputational concerns or policy choices.¹ This paper does not restate that framework. It applies it to one artefact and therefore begins only with the provisions that anchor the analysis.²

Method and scope. This paper uses a legal-technical design-analysis method. It begins from existing international humanitarian law obligations concerning persons hors de combat and verification, identifies the operational difficulty that arises when an engagement is conducted at machine speed, and assesses one

¹ Giovanni Nardacci, Lawful Operational Safeguards in AI Systems: Surrender Recognition, Compliance Architecture, and International Humanitarian Law (Version 9, September 2026), Zenodo DOI 10.5281/zenodo.21932439; SSRN Abstract ID 6821838 [hereinafter “Paper I”]. The canonical definition of “lawful operational safeguard” appears in Section VI.A; the feasibility criteria in Section V.A; and the treatment of failure modes in Section VII.

² Protocol Additional to the Geneva Conventions of 12 August 1949, and relating to the Protection of Victims of International Armed Conflicts (Protocol I), 8 June 1977, 1125 UNTS 3 [hereinafter “AP I”]. All treaty provisions quoted in this paper were checked verbatim against the official text published in the ICRC IHL Treaties Database, ihl-databases.icrc.org.

candidate technical referent against previously stated feasibility criteria. The analysis is intentionally *ex ante*. Its purpose is to determine whether the proposal is specified closely enough to be tested and reviewed. It does not ask whether the proposal has already been shown to work. No experimental result, deployment claim or legal status is inferred from the existence of the specification. Treaty provisions are used as normative anchors. Engineering values inherited from the 2025 file are treated as candidate parameters. Where the legacy specification is incomplete, the later HFA technical notes are used to define reproducible test conventions without retroactively converting those conventions into requirements of that file.

The underlying duty to protect persons *hors de combat* is customary (ICRC Rule 47; Hague Regulations 1907, Art. 23(c)) and binds all parties to an armed conflict. Its clearest treaty formulation, used here to anchor the technical analysis of machine recognition, appears in Additional Protocol I:

1. A person who is recognised or who, in the circumstances, should be recognised to be *hors de combat* shall not be made the object of attack.
2. A person is *hors de combat* if: (a) he is in the power of an adverse Party; (b) he clearly expresses an intention to surrender; or (c) he has been rendered unconscious or is otherwise incapacitated by wounds or sickness, and therefore is incapable of defending himself; provided that in any of these cases he abstains from any hostile act and does not attempt to escape.³

Article 57(2)(a)(i) requires those who plan or decide upon an attack to “do everything feasible to verify that the objectives to be attacked are neither civilians nor civilian objects...”⁴

Three features of these provisions carry the argument. First, they are already in force. Both have been treaty law since 1977 (in force from 1978), and the underlying rule is older customary law. Neither depends for its validity on any particular stage of weapons technology. That remains true for the systems considered here and is not disputed in the forum in which they are discussed. The Guiding Principles affirmed by the Group of Governmental Experts state that international humanitarian law continues to apply fully to all weapons systems, including the potential development and use of lethal autonomous weapons systems.⁵ That principle settles the applicability of the law; it does not answer the questions raised here, which begin after that point. Second, the obligations are means-neutral because they attach to the party conducting the attack, whatever instrument is used. Third, and crucially for present purposes, Article 41(2)(b) presupposes an act of communication. A person is *hors de combat* under that limb because he or she clearly expresses an intention to surrender. The rule protects an expression, and an expression is effective only if its addressee can read it. The 1987 ICRC Commentary already recognised that no single form is prescribed. Formal surrender is not always realistically possible, and the intention must be conveyed by whatever means the circumstances permit.⁶

The effectiveness gap appears here. Where the addressee is human, raised hands, a raised white cloth or a discarded weapon can be understood within the time taken by a human decision. Where the addressee is a system that detects, classifies and engages within milliseconds, there is no assurance that a human-legible gesture will be read before the engagement decision is executed. Nor is there presently a shared machine-readable code through which surrender can be expressed and recognised at that speed. The flexibility recognised by the Commentary is itself dependent on the addressee. A human can interpret without a predefined code because the human supplies meaning to the raised white cloth. A machine can perceive only what has been defined for it in advance; an expression for which no class has been defined is not merely misread — it may not be read at all. What is missing at machine speed is therefore not only time but also a form that machine perception can be built around. In practical terms, that means a fixed, published referent. This difficulty is independently documented. The cited literature identifies the variability of surrender expressions, their dependence on context and their performance under adversarial conditions as obstacles to reliable surrender recognition by autonomous systems.⁷ The result is not a normative gap. The rule remains

³ AP I, Art. 41(1)–(2).

⁴ AP I, Art. 57(2)(a)(i).

⁵ Guiding Principles affirmed by the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems, Annex IV, CCW/GGE.1/2019/3 (2019), Principle (a): international humanitarian law continues to apply fully to all weapons systems, including the potential development and use of lethal autonomous weapons systems.

⁶ Yves Sandoz, Christophe Swinarski and Bruno Zimmermann (eds.), *Commentary on the Additional Protocols of 8 June 1977 to the Geneva Conventions of 12 August 1949* (ICRC/Martinus Nijhoff, 1987), para. 1612 (on Art. 41: a formal surrender is not always realistically possible; the expression of an intention to surrender depends on the means available in the circumstances).

⁷ The difficulty is independently documented. See Robert Sparrow, “Twenty Seconds to Comply: Autonomous Weapon Systems and the Recognition of Surrender”, 91 *International Law Studies* 699–728 (2015) (U.S. Naval War College), analysing why reliable surrender recognition by autonomous weapon systems remains unresolved and why systems lacking it are likely to remain controversial; and ICRC, *ICRC Position on Autonomous Weapon Systems* (12 May 2021), with background paper, observing that anti-personnel autonomous weapon systems incapable of recognising an enemy’s clear expression of an intention to surrender, and refraining from engagement, raise serious IHL concerns — and recommending constraints accordingly. On the content and practical conditions of the rule of surrender

intact. What is absent is a means of giving it practical effect. The same structure — an existing obligation resting upon an implicit assumption about human reaction time that a class of machines silently removes — was examined in the author’s second paper in the context of civilian machine-safety law.⁸

Position within the cited literature. The materials cited in this paper address several related but distinct parts of the problem. They cover the legal content of surrender and hors de combat protection, the difficulty of reliable surrender recognition by autonomous systems, and the broader international humanitarian law constraints applicable to autonomous weapon systems. This paper does not replace behavioural recognition and does not argue that a machine-readable referent is legally required. Its narrower contribution is to place a concrete, public and technically specified referent within that problem space and to ask what becomes measurable once such an artefact exists. The contribution is therefore not evidence that HF SIGNAL 01 succeeds, but a testable object and an explicit separation between legal protection, signal semantics and measurable recognition performance.

No new rule follows from this, and the proposal is not — yet — a standard. The obligation is unconditional. Any path producing the same effect stands on the same footing; a party that follows none is not choosing a different method, it is not performing the obligation. The principal response paths include (1) restoring meaningful human decision time through genuine supervision or doctrinal limits on autonomy; (2) training machine perception to recognise existing human expressions of surrender through validated behavioural recognition; (3) introducing a shared machine-readable referent — a public, dated and technically specified code that machine perception can be built to read; or (4) constraining the system’s target sets, environments, speed or operating modes so that legally protected changes of status can still affect conduct before force becomes irreversible. These paths may be combined. The paths are not, however, equally available today. The difference is one of state of the art rather than principle. The first turns on doctrine and command decisions rather than technical development and, where genuinely adopted, removes the machine-speed gap that motivates this paper. The second is legitimate in principle and is not excluded here, but it remains an open research programme rather than an available general solution. The literature reviewed here identifies neither a generally accepted specification of the human behaviours to be recognised nor a validated general-purpose recogniser, and the difficulty of variation across cultures and circumstances, especially under adversarial conditions, is well documented. The fourth path restricts the capability itself — its targets, environment, speed, duration or operating mode. It constitutes compliance where the restriction removes the problematic circumstance from the operational envelope; where it merely reduces the chance that the circumstance will arise, it mitigates risk without providing a means to register a change of condition. A narrower geographical perimeter, for example, may make surrender within that perimeter less likely, but it neither excludes the possibility nor enables the system to recognise it. That path is legitimate, but it may impose substantial operational cost and may therefore be difficult to sustain under pressure. For present purposes, the third path has three properties that make it especially amenable to ex ante specification, testing and legal review. The case for examining it rests on those properties rather than on the merits of any particular signal. Two propositions must remain distinct. First, a person recognised as hors de combat may not be made the object of attack; that is the customary rule on which this analysis rests. It does not follow that any particular system must therefore be built to recognise HF SIGNAL 01 or any other specific referent. The second proposition requires a further step. Where a mode of employment makes it reasonably foreseeable that persons may acquire a protected condition during an interval in which the engagement can still be altered, the question is what a party must do to avoid making that protection systematically ineffective. This paper asks what a fixed referent would make examinable if that question were posed. A fixed referent does not require contextual interpretation of an open-ended human behavioural repertoire. Behavioural recognition does; the first and fourth paths depend instead on judgements made elsewhere about the reality of supervision or the adequacy of restriction. A fixed referent asks a narrower question. The inquiry is simply whether a defined signature is present. That is a matter of measurement rather than understanding. It also creates a representable state — a condition that the system can hold, record and act upon as a distinct value rather than one it must infer. A system unable to represent a protected condition cannot register its arrival, cannot log it and cannot act upon it; the condition may exist in fact while remaining absent from every process governing the engagement. A fixed referent is also particularly suited to direct quantitative verification before an encounter. It can be measured in advance whether a system

itself, see Russell Buchan, “The Rule of Surrender in International Humanitarian Law”, (2018) 51(1) *Israel Law Review* 3–27; on IHL compliance questions raised by autonomous weapon systems generally, see Laura Bruun, Marta Bo and Netta Goussac, *Compliance with International Humanitarian Law in the Development and Use of Autonomous Weapon Systems: What does IHL Permit, Prohibit and Require?* (SIPRI, March 2023).

⁸ Giovanni Nardacci, *Systemic Arrestability: A Temporal Criterion for Machine Safety*, Zenodo, DOI 10.5281/zenodo.20837150; SSRN Abstract ID 6923461 [hereinafter “Paper II”]. Paper II applies the same method to civilian machine-safety law: an existing obligation rests upon an implicit temporal assumption that a class of machines removes.

detects the signature, at what distance, with what latency and with what error rate. Supervision, behavioural recognition and operational restriction are less reducible to a bounded pre-encounter test and depend more heavily on context, generalisation and operating conditions. Finally, a fixed referent need not require the prior removal of autonomy or a categorical restriction of intended employment. Instead, it adds a detector rather than removing a capability and can therefore be adopted without first making that policy choice, though not without operational cost. A legal-review consequence follows. Because a referent can be specified in advance, its absence from a system can also be identified in advance; that makes the question examinable during procurement rather than reconstructible only after an engagement. A further consequence concerns the systems considered here. The first path assumes a human in the engagement loop whose decision time is restored; the fourth narrows permitted use. Where a system is intended to operate without such a human and without narrowing its intended employment, the capacity to register a protected condition has nowhere to reside except within the system. The two relevant system-internal candidates are then behavioural recognition and a fixed machine-readable referent. The distinction between them is the one already identified. The detection performance of a fixed referent can be specified and quantitatively tested against a defined signal in advance, whereas the generalisation of a behavioural recogniser cannot be established in the same bounded way.

HF SIGNAL 01, described in the next Section, is a candidate for the third path. It is not presented as the solution or as superior to the alternatives. It is a concrete attempt to define a machine-readable referent with enough technical detail to permit experimental scrutiny, and it is offered for scrutiny — including hostile scrutiny — rather than for adoption.

2. The Protocol: HF SIGNAL 01 Technical Specification

2.1 Dual-channel architecture

HF SIGNAL 01 is a dual-channel signal. The full specification is the protocol issued on 24 April 2025 and preserved in an integrity-protected file; it is summarised in the May 2026 Technical Note,⁹ which is the Association's current statement of the specification.¹⁰ Recognition is valid only when both channels have been confirmed. The two channels are described below.

Optical channel. A pulsating white light (colour code #FFFFFF) at a pulse frequency of 7 Hz, with a uniform pulse pattern, a recommended source brightness of at least 300 lux, a nominal visibility range of 100 m and infrared compatibility. The hexadecimal value is a digital colour description. Because sRGB uses D65 as its reference white, HFA-TN-03 uses D65 as the reporting reference for physical chromaticity without treating any chromaticity tolerance as an existing requirement of the 2025 file. “Infrared compatibility” is retained as an implementation objective inherited from that file, but it is not presently a conformance parameter because no infrared wavelength, radiant-output level or receiver criterion is specified. A periodic 7 Hz signature is intentionally simple because it lies well below the Nyquist limit of ordinary camera frame rates. This means that the temporal pattern is in principle sampleable by common sensors without requiring a specialised detector. Whether a particular camera resolves the signature at a particular distance is a different question, governed by receiving optics and ambient conditions rather than sampling rate. Published optical-camera-communication literature indicates that detection at distances of the order of 100 m has generally required dedicated optics.

Radio channel. A continuous FM tone at 144.0 MHz — within the amateur VHF band — with a bandwidth of 25 kHz and a 1,000 Hz tone; the specification states a detection threshold of -85 dBm and recommends threshold crossing as the detection method.

The redundancy rule is explicit in the specification (“dual_channel_required”). A system must confirm both the optical and radio signatures before treating the signal as present. The conjunction is the protocol's principal defence against accidental activation. Emission takes place in ten-second cycles and may be triggered manually or automatically. The ten-second duration is an emission-policy value inherited from the 2025 file, not a feature required to identify the machine-readable signature. Its suitability is therefore a separate operational and safety question rather than a prerequisite for recognition testing.

Design rationale. The choices embodied in the specification follow from constraints implicit in its purpose. The controlling constraint lies on the recognising side. The signal must be readable by sensor classes already

⁹ Human Flag Association, HF SIGNAL 01 — Technical Note on AI Safeguards and International Humanitarian Law (May 2026), HF_SIGNAL_01_Technical_Note_May2026.md, in the public repository github.com/nardaxx/hf-signal-01. The Technical Note summarises the specification; the complete parameter set is contained in the integrity-protected file cited in the following note.

¹⁰ Human Flag Association, HF_SIGNAL_01_SOFTWARE_FULL_EXTENDED.txt (issued 24 April 2025), in the repository cited supra note 9; integrity protected by SHA-256 hash 3a7de50d3cec9843aae11fcfe1c295f7ffb8e7eac3cbfc5851a43b509812182e and OpenTimestamps certification (HF_SIGNAL_01_SOFTWARE_FULL_EXTENDED.txt.ots). All parameter values quoted in this Section are drawn from that file and from the Technical Note, supra note 9.

commonplace on autonomous platforms — cameras and general-purpose radio receivers. That excludes QR-type visual codes, RFID, Bluetooth and encrypted transponders. A second constraint lies on the emitting side — hardware should be minimal, inexpensive and powered — and a third is redundancy across two independent physical channels. Within those constraints, the specific numerical values reflect engineering judgement. Their value at this stage is that they are fixed and therefore testable.

Earlier design iterations appear in the public history of the Association's repositories. As clarified in the accompanying Architecture Note, however, the protocol now comprises two alternative modalities. The first is the active electronic dual-channel specification described here. The second is a passive visual modality consisting of a white disc on a red field. The passive modality is a current part of the protocol, not a superseded design stage; it is the option available to a person without powered equipment. The two modalities are alternative expressions of the same protective meaning.

2.2 Operation in practice

The specification goes beyond the physical signature. It defines when the signal is emitted, by whom, and what a recognising system is expected to do. Those operational provisions matter to the legal analysis because they present the protocol as a complete communicative act rather than merely a waveform.

Activation. Three activation modes are defined. The first is manual activation. A person facing a perceived imminent threat deliberately activates the signal, with a required confirmation step so that emission is a conscious act. The second is proximity-triggered activation. It occurs when an autonomous system or unidentified threat is detected within a nominal radius of 50 metres, using infrared, visual, motion or RFID sensing on the emitter side. The third is pre-programmed periodic broadcast — at a nominal interval of fifteen minutes — during exposure to a known high-risk environment, for example while moving through an area in which autonomous or semi-autonomous systems are known to operate. The 50-metre and fifteen-minute values are activation-policy defaults, not elements of the optical or radio signature presented to the recognising system, and the feasibility argument in this paper does not depend on their numerical values. If retained, they require operational validation, but they need not be validated in order to determine whether the signal itself can be recognised. The specification identifies a human operator, an unarmed drone and a civilian device as intended emitters, and a combat zone, an urban-conflict area and an automated-surveillance area as intended environments. It also imposes a condition worth emphasising. Activation must occur in good faith and under perceived threat. The signal is an expression of protected status, not a general-purpose beacon, and the specification treats it accordingly.

Expected response. On the recognising side, the specification defines what a system that has confirmed both channels is expected to do. It should immediately cease aggressive behaviour, initiate a non-hostile identification procedure, and give priority to the preservation of human life. Two qualifications must remain in view. First, these are specification-side expectations — statements about what compliant recognition logic is expected to do — not descriptions of implemented behaviour; no system implementing them currently exists (Section 3). Second, they map directly onto the structure of Article 41. Ending the engagement gives operational effect to the requirement that the person “shall not be made the object of attack”, while a subsequent identification process carries out the verification required by Article 57(2)(a)(i). The expected-response clause is therefore the point at which the technical document and the treaty text meet.

Emitter hardware. The emitter side is deliberately low-technology and consists of a white LED or xenon source capable of pulsing at 7 Hz, together with a low-power VHF transmitter capable of a continuous 1,000 Hz FM tone at 144.0 MHz. The specification requires no bespoke equipment. That is a deliberate design choice because a standard that only well-resourced actors could emit would defeat its own purpose. Even so, the protocol assumes that a functioning, powered emitter is available at the moment it is needed, and that assumption is itself a limit of the protocol, addressed in Section 5.

2.2a Legal classification of activation modes

The legal significance of the signal depends on how it is activated. Manual activation by a person, or by a device acting upon that person's deliberate and contemporaneous instruction, may provide evidence of an expressed intention to surrender within Article 41(2)(b) of Additional Protocol I, provided the surrounding circumstances confirm that the expression is genuine and the person refrains from hostile acts.

Automatic, proximity-triggered or periodic activation does not by itself establish such an intention and is not treated by this specification as an act of surrender. It functions only as a precautionary protective alert requiring temporary suspension of engagement, verification by a human or by the system, and assessment of the surrounding circumstances. Likewise, signals originating from unmanned devices do not confer *hors de combat*

status on an object; they may communicate on behalf of identifiable persons or trigger a precautionary response.

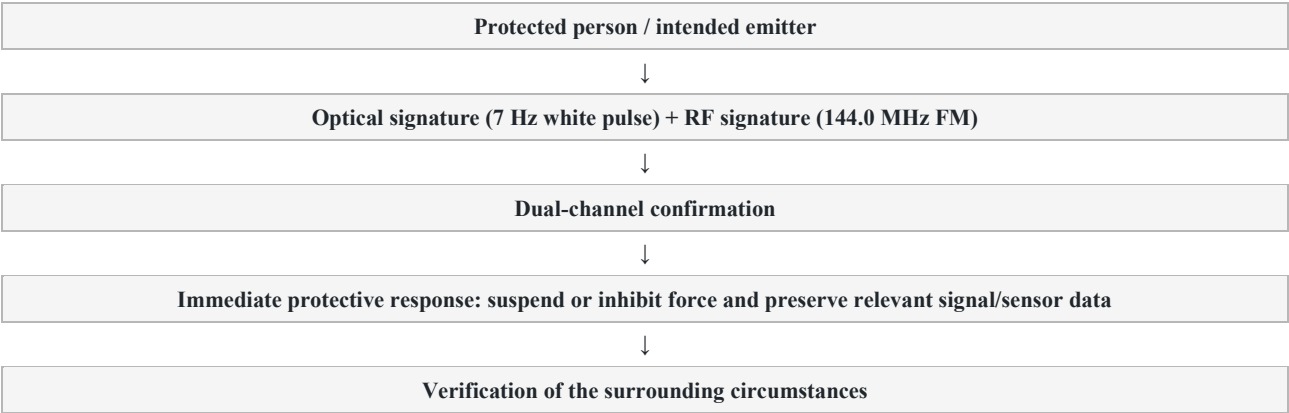
That distinction is not a flaw in the specification but a necessary consequence of the underlying law. A machine-readable signal can express an intention only where a human intention is in fact being expressed. Where it is not, the signal retains a precautionary protective function but does not itself engage the specific protections associated with surrender or hors de combat status.

2.3 Direction of the signal: emitter and addressee

Surrender involves two roles that are easy to conflate. One is the person who surrenders. The other is the actor who must recognise the surrender. Article 41 places the legal burden on the latter — the attacker must not attack a person who “should be recognised” as hors de combat. HF SIGNAL 01 addresses the machine instantiation of that second role by giving the first role a code that the second can read. The protected person emits the signal in order to be recognised by the system. Its direction is therefore from protected person to machine. It is a recognition signal, not a targeting instrument.

Two clarifications follow. First, the signal does not confer status by itself. A person is hors de combat because the conditions in Article 41(2) are met, however that status is expressed. HF SIGNAL 01 adds a machine-legible means of expression; it is not an independent source of protection. The design rule that operationalises this point is stated once here and applies throughout. Under that rule, compliant recognition logic treats a confirmed signal as sufficient to trigger an immediate protective response — suspension or inhibition of force, preservation of the signal and relevant sensor data, and verification of the surrounding circumstances — but not, by itself, as conclusive proof of hors de combat status. The signal is never a necessary condition of protection. Its absence supports no adverse inference, and the full range of human-legible expressions of surrender retains its legal force. Any implementation that reversed this logic — treating non-emission as an indication of targetability — would violate Article 41 rather than implement it. Second, HF SIGNAL 01 is not a recognised emblem, sign or signal for the purposes of Article 38 of Additional Protocol I,¹¹ and this paper claims no such status for it. It is a candidate technical standard whose legal significance derives entirely from the pre-existing obligation it would help make practicable.

Figure 1. Conceptual functional flow of HF SIGNAL 01



Conceptual flow only. Confirmation triggers a protective response and verification; it does not by itself establish hors de combat status.

2.4 The ethical clause in the 2025 file: scope and legal character

The integrity-protected specification contains an ethical clause that a careful reader will, and should, find. Its “forbidden_use” list includes “military targeting systems” and “weapons”, while the accompanying declaration states that use of the signal “for military targeting, suppression, deception, or monetisation constitutes a violation of international humanitarian ethics”.¹² Read in isolation, this might appear to prohibit the very activity discussed in this paper — processing the signal within weapon systems. The apparent tension resolves on two matters. The first is the direction in which the signal is used. The second is the legal character of such clauses.

The prohibition concerns offensive instrumentalisation of the signal. This includes feigning it in order to induce reliance and then strike, exploiting its emission to locate, identify and engage the emitter, or feeding it into an engagement chain as targeting input. In each case the signal is turned against the person it exists to protect. This paper addresses the opposite direction — recognition in order to abstain. Recognition-for-abstention does not

¹¹ AP I, Art. 38(1) (prohibition of improper use of recognised emblems, signs and signals).

¹² HF_SIGNAL_01_SOFTWARE_FULL_EXTENDED.txt, supra note 10, ethical_clause.

“use the signal for targeting”; it makes the signal legible, which is the signal’s purpose, just as teaching a soldier to recognise a white flag is not a military use of the white flag. The distinction follows the two roles identified in Section 2.3. The prohibition protects the person surrendering, while recognition capability is required of the actor who must recognise surrender.

As to legal character, the clause forms part of the April 2025 file, which is integrity-protected and therefore cannot be amended, and reflects the protective intent present when that file was issued. The May 2026 Technical Note, which summarises the specification, does not repeat the clause. That is not an oversight; it clarifies what such a clause can and cannot do. In the case of a signal, a prohibition written by its issuer is declaratory rather than constitutive. A white flag carries no licence terms, and its protection against abuse derives from the law itself — including the rules on perfidy and improper use of signs and signals — regardless of any condition the issuer might attach and regardless of the signal’s status. The 2025 clause therefore adds emphasis rather than a new obligation, and the analysis in this paper does not depend on it. Its intended reading is nonetheless stated plainly. Implementing recognition of HF SIGNAL 01 in order to withhold or break off engagement is an intended use of the specification. Processing the signal in a way that contributes to selecting or engaging the emitter as a target is abuse — condemned by the clause and, where the elements of Article 37 are satisfied, prohibited by law. Separately, on an entirely different plane, the specification document is distributed under CC BY-NC-ND 4.0, free for humanitarian and civilian use, while institutional use is subject to a written licence.¹³ That is a sustainability model for the issuing association and has no bearing on the international-humanitarian-law analysis of signal recognition.

2.5 Precedent within the treaty architecture: distinctive signals

A reader encountering a protective light-plus-radio signal for the first time might assume that the concept itself is novel. It is not. Additional Protocol I defines the “distinctive signal” in Article 8(m) as “any signal or message specified for the identification exclusively of medical units or transports” in Chapter III of Annex I, and Article 18(5)–(6) authorises Parties to use such signals to identify medical units and transports. Annex I, as amended, specifies these signals in technical terms. They include a flashing blue light with a recommended flash rate of sixty to one hundred flashes per minute, a radiocommunication signal, and electronic means of identification.¹⁴ In other words, the architecture of the Protocol already contains protection-relevant communication in the form of technically defined light and radio signatures designed to be recognised by instruments as well as by human observers, with parameters set out in a technical annex and maintained over time. Annex I was revised in 1993 precisely so that its specifications could remain current.

The comparison must be drawn carefully because it proves only a limited point. HF SIGNAL 01 is not a distinctive signal under Annex I, has no status by virtue of that Annex and borrows none of its protection, consistently with the position taken on Article 38 in Section 2.3. The comparison establishes only that the kind of object HF SIGNAL 01 proposes to become is already familiar to the Protocol. A specified pulse frequency for a light and a specified carrier for a radio emission, used to make a protective rule practically operable, are not alien to international humanitarian law; the same technical approach is already used within the Conventions’ machinery for the medical function, applied here instead to the expression of surrender at machine speed. Everything the Annex I signals possess and HF SIGNAL 01 presently lacks — State negotiation, treaty authorisation and institutional maintenance — describes the distance still to be travelled. It is precisely the gap that the invitation in Section 6 asks competent institutions to consider. Annex I also shows the institutional route by which such specifications evolve through proposal, technical review, amendment and maintenance by competent bodies. That is the route Section 6 proposes for this candidate.

3. Feasibility Mapping against the Criteria of Paper I, Section V.A

Paper I sets out feasibility criteria that any candidate lawful operational safeguard must ultimately satisfy. They cover technology readiness, latency thresholds, annotated datasets and independent validation.¹⁵ A paper that advanced a candidate and then exempted it from the author’s own criteria would not merit serious

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¹⁴ AP I, Arts. 8(m) and 18(5)–(6); Annex I to AP I, Regulations concerning identification (as amended on 30 November 1993), Chapter III (light signal, radio signal, electronic identification).

¹⁵ Paper I, *supra* note 1, Section V.A.

consideration. This Section therefore assesses HF SIGNAL 01 against each criterion without changing them. The table provides the summary; the paragraphs that follow explain the assessment.

Table 1. Feasibility mapping against the criteria of Paper I, Section V.A

Criterion (Paper I, §V.A)	What the criterion requires	Status of HF SIGNAL 01 (July 2026)
Technology readiness	Demonstrated maturity of the recognition capability, at least to the level of analytical or experimental proof of concept.	TRL 2. A candidate parameter set exists and permits experimental characterisation; no proof of concept, experiment or measurement exists.
Latency thresholds	Measured detection latency compatible with the engagement timescale of the systems concerned.	No latency measurement exists. None is claimed.
Annotated datasets	Curated positive and negative samples across operational conditions, sufficient to train and evaluate detectors.	None exist. The specification's reference to a YOLOv8-class model is a recommendation; no model has been trained.
Independent validation	Testing by an actor other than the proponent — a laboratory, standards body or equivalent.	None has been performed. Deposit and timestamp establish provenance, not validity (Section 4).

Technology readiness. HF SIGNAL 01 stands at TRL 2, which corresponds to a technology concept that has been formulated.¹⁶ No prototype has been built to the specification, no field or laboratory trial has been conducted, and no performance figure has been produced. TRL 3 requires analytical or experimental proof of concept; neither exists, and neither is claimed here. Within TRL 2, one feature deserves separate mention. The specification fixes a candidate parameter set in enough detail to allow initial experimental characterisation. Pulse frequency, colour code, brightness, visibility range, carrier frequency, bandwidth, tone frequency, detection threshold and redundancy rule are all stated numerically, making the protocol sufficiently determinate for third-party testing to begin. Several values nevertheless require metrological normalisation before independently built emitters can be treated as equivalent. The colour code identifies white in a digital colour space rather than the chromaticity of a physical source; the brightness value lacks a declared measurement geometry; and the optical duty cycle is not fixed. The RF side contains an analogous gap. A stated 25 kHz bandwidth and 1,000 Hz tone do not uniquely define an analogue FM waveform without a declared frequency deviation or equivalent modulation parameter, while the -85 dBm threshold needs a defined receiver-input reference. HFA-TN-03 and HFA-TN-06 therefore provide reproducible test conventions for those quantities without retroactively turning those conventions into requirements of the 2025 file. Specification sufficient for testing is a genuine strength at this stage — but it is neither completeness nor evidence of performance.

Latency. The criterion asks whether detection can occur within the decision timescale of the systems concerned. No latency has yet been measured for HF SIGNAL 01. A plausibility argument can be made. A 7 Hz signature can in principle be confirmed within a few pulse periods, that is, within several hundred milliseconds of observation, while threshold detection of a continuous -85 dBm tone is routine in software-defined radio. Plausibility is not measurement, however, and independent testing exists precisely to establish the difference. This paper makes no latency claim.

Annotated datasets. No dataset exists. There is no positive and negative optical corpus spanning distance, weather, illumination, motion and occlusion, no library of radio captures across interference and multipath, and no adversarial sample set. Consequently, no detector has been trained, and the protocol's false-positive and false-negative rates — figures on which any eventual deployment argument would depend — are unknown. Dataset construction is the single most consequential open task because every other criterion ultimately depends upon it.

Independent validation. None has yet taken place. No university laboratory, standards organisation or other third party has examined the protocol. The documentary apparatus described in Section 4 — deposit, hash and timestamp — establishes that the specification existed in this exact form at a certified date. It says nothing about whether the specification works. Those are different kinds of evidence and must not be confused.

Measured against its own framework, HF SIGNAL 01 presently satisfies only the precondition assumed by that framework's criteria. A fixed, public candidate exists and is specified closely enough to be tested. That qualification matters. The active electronic modality is defined at a level that permits direct testing, subject to the metrological normalisation described above, while the passive visual modality is architecturally defined but still awaits normative parameterisation of the values identified in Section 5. Everything else remains to be done. So that “everything else” is not left vague, the validation task can itself be specified.

¹⁶ The Technology Readiness Level (TRL) scale is used here in the sense of ISO 16290:2013, Space systems — Definition of the Technology Readiness Levels (TRLs) and their criteria of assessment, whose definitions are in general engineering use. TRL 2 corresponds to a technology concept formulated; TRL 3 requires analytical or experimental proof of concept.

What validation would require. For the optical channel, validation would require a factorial capture campaign producing positive samples across distances up to and beyond the nominal 100 m, across lighting regimes, weather conditions, emitter motion and partial occlusion; a negative corpus of confusable periodic light sources recorded under the same conditions; and, where relevant, deliberate imitations. HFA-TN-03 reduces that work to a preregistered set of receiver classes, measurements and statistical outputs. Initial radio validation need not begin with radiated transmission. HFA-TN-06 uses conducted 50 Ω bench testing with a signal generator or software-defined radio and calibrated attenuation, allowing the legacy -85 dBm threshold, detection latency and false-positive behaviour to be tested before transmitter power, antenna gain or radiated range are fixed and without unnecessary occupation of the amateur band. Radiated trials belong to a later stage and require an authorised frequency arrangement together with a specified emitter. HFA-TN-05 then measures the conjunction directly. Reference detectors would produce the quantities on which validation turns. These include false-positive and false-negative behaviour and time-to-confirmation distributions for each channel and for the required conjunction of both. Replication by a party other than the proponent would complete the move from TRL 2 to demonstrated proof of concept. None of this requires cooperation from a weapons manufacturer or a State; it is ordinary machine-perception and communications testing that a university or competent test laboratory can perform.

A further open question belongs to any future standardisation process rather than to this paper. The protocol's radio channel sits within an allocated amateur band, and that choice merits critical examination. Spectrum regulation varies by jurisdiction; the band carries legitimate traffic and may be congested; in a conflict zone it may be deliberately jammed or unavailable; and transmission by unlicensed persons raises regulatory questions requiring coordination with the competent spectrum authorities. The designated carrier creates two additional issues of different kinds. The first issue is regulatory. Because 144.0 MHz is the lower edge of the amateur allocation, an emission 25 kHz wide and centred on that frequency would place roughly half of its bandwidth below 144 MHz, in spectrum allocated to other services, which differ by ITU Region. The second issue concerns coordination rather than law. Regional IARU band plans reserve the lower portion of the 2-metre band for narrow-band modes. In Region 1, 144.000–144.135 MHz is reserved exclusively for telegraphy, so a wideband FM emission would not conform to those plans even if it remained within the allocation; FM telephony segments lie higher in the band. Neither issue affects detectability and neither is resolved here. Both show that the carrier is a value for coordination, and that moving the carrier clear of the band edge would remove the first problem without changing the rest of the specification. A standards body would also be the proper forum to decide whether 144.0 MHz should remain the designated carrier or whether a dedicated protected allocation — the treatment given to Annex I signals for the medical function — would be justified. The specification fixes a testable value; it does not close that discussion.

4. Provenance and Documentary Integrity

The chronology of the specification is short and fully documented. The complete protocol was finalised on 24 April 2025. It was deposited with the Legal Services of the State Council of the Canton of Ticino on 28 April 2025¹⁷ and published in a public repository in May 2025. The protocol file is protected by SHA-256 digest 3a7de50d3cec9843aae11fcfe1c295f7ffb8e7eac3cbfc5851a43b509812182e and by an OpenTimestamps certificate anchoring that digest in the Bitcoin blockchain.¹⁸ A reader can therefore verify, without trusting the issuing association, that the file examined today is byte-for-byte the file that existed at the certified date. The May 2026 Technical Note summarises the specification; the parameter values are those of the integrity-protected file.

The purpose of this documentary apparatus must be stated precisely because it is easy to overstate. Deposit, hash and timestamp establish two things. First, they establish intellectual priority, meaning that the specification with these exact parameters was publicly in existence from these dates. Second, they establish integrity against later alteration, meaning that it cannot have been silently changed since. They establish nothing more. In particular, as Section 3 has already made clear, they are not evidence of technical validity and are not presented as such.

The record can also be verified without relying on its author. Any reader may recalculate the protocol file's SHA-256 digest and compare it with the stated value; any reader may verify the OpenTimestamps proof against the Bitcoin blockchain using freely available open-source tools; and the cantonal deposit provides an institutional record independent of the Association's own infrastructure. Each element was selected so that it

¹⁷ Deposit with the Legal Services of the State Council of the Canton of Ticino, Bellinzona, 28 April 2025.

¹⁸ OpenTimestamps (opentimestamps.org) anchors a file's SHA-256 digest in the Bitcoin blockchain, allowing the existence of the exact file content at the certified date to be verified independently of the issuing party.

can be placed before a review body, a standardisation process or an intergovernmental forum and examined on its own terms without depending upon assertions by the issuing association.

What documented existence changes is the description of the state of the art. So long as no candidate standard exists, the absence of surrender recognition in an autonomous system can be characterised as an inherent technical limitation because there was nothing specified to recognise, and the binding obligation in Article 41 had no validated, operationally available means of implementation at machine speed. Once a candidate standard is publicly available, dated and integrity-protected, that characterisation is no longer a complete account of the facts. Examining the candidate, testing it, adopting it, improving it or rejecting it on stated grounds becomes a decision — one that may be made well or badly, but one that can be documented and reviewed. The existence of a specified candidate therefore shifts the assessment of non-recognition from an unavoidable technical limitation to a choice capable of legal scrutiny. One clarification prevents this proposition from being overstated. A reasoned conclusion, reached after examination, that the candidate is technically unsound would itself constitute the examination contemplated by this argument. The record would then show a standard proposed, tested and rejected on stated grounds, which is a legitimate outcome. What the record rules out is only the assertion that there was nothing to examine.

Article 36 of Additional Protocol I supplies the established model for that shift. A High Contracting Party studying, developing, acquiring or adopting a new weapon, means or method of warfare “is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited” by the Protocol or another applicable rule.¹⁹ The chain from that text to the argument here is short, and each link lies in the treaty. For States Parties, Article 36 requires the reviewing State to determine whether employment would be prohibited; Article 41 prohibits attacks upon a person who should be recognised as *hors de combat*. The system’s capacity to make that recognition therefore falls within the subject matter of the review rather than forming an optional appendix to it. Any determination of what is feasible must also be made by reference to the state of the art, including material publicly available at the time of review. Nothing in this chain requires Article 36 to name third-party candidate safeguards, and this paper does not read it that way. The treaty text mentions no such candidates and no such requirement is asserted. The narrower point is that a review asking whether machine-speed employment would violate Article 41 has a defined question before it, and the publicly available material bearing on that question now includes a candidate means specified in testable form. This paper adds only the fact and its date. Such a candidate has been publicly available since May 2025, and in summarised form since May 2026. What follows from that fact, and in which forum, is not for this paper to determine.

5. Limits

This Section identifies what the protocol does not resolve. The failure modes are those addressed in general terms in Paper I, Section VII,²⁰ made concrete here for this specification, together with the condition that governs everything else. Adoption is presently voluntary.

False positives. A 7 Hz white light may arise incidentally — from emergency strobes, damaged luminaires or reflections of periodic sources — and 144.0 MHz sits in an allocated amateur band carrying legitimate traffic. The dual-channel conjunction is intended to make coincidence rare, but how rare is an empirical question, and the datasets needed to answer it do not yet exist. The operational cost of a false positive is unnecessary abstention. A protocol with a high false-positive rate would create precisely the pressure to ignore it that a standard ought to prevent. The number matters, and it is currently unknown.

False negatives. Emission can fail through equipment damage, loss of power, terrain masking or jamming, and detection can fail even where emission succeeds. Two consequences must be reflected in any recognition logic. First, absence of the signal carries no adverse inference. Article 41 protects any person who clearly expresses an intention to surrender by whatever means, and HF SIGNAL 01 adds a machine-legible means without narrowing the class of valid expressions. A system must never be configured to treat non-emission as an indication of hostility. The protocol is a floor for legibility, not a filter on protection. Second, where a detected signal is later accompanied by hostile acts, Article 41’s own proviso applies; protection is conditional upon abstention from hostile acts.²¹ The protocol changes nothing in that respect.

Adversarial exploitation. The signal can be feigned. Existing law already addresses the gravest form of that conduct. Article 37(1) prohibits killing, injuring or capturing an adversary by resort to perfidy, defined as “[a]cts inviting the confidence of an adversary to lead him to believe that he is entitled to, or is obliged to

¹⁹ AP I, Art. 36. On the reading of Art. 36 adopted here — that the review obligation makes the treatment of known and available means of compliance examinable — see Paper I, *supra* note 1.

²⁰ Paper I, *supra* note 1, Section VII.

²¹ AP I, Art. 41(2), final proviso.

accord, protection under the rules of international law applicable in armed conflict, with intent to betray that confidence", and lists as its first example "the feigning of an intent to negotiate under a flag of truce or of a surrender".²² Falsely emitting HF SIGNAL 01 in order to induce reliance and then kill, injure or capture engages those elements irrespective of the signal's status. The perfidy rule attaches to the feigning of surrender itself, whatever code is used to feign it. The protocol therefore creates neither a new protection nor a new vulnerability in perfidy terms. Two qualifications are necessary. Perfidy *stricto sensu* requires the result element of killing, injuring or capturing; false emission used solely to avoid engagement does not satisfy that definition, though it degrades the signal's reliability for everyone and may engage other rules on improper use of signs and signals. Nor does this analysis assimilate HF SIGNAL 01 to the recognised emblems protected by Article 38. It is not one of them, and its misuse is not misuse of those emblems. The technical answer ultimately demanded by adversarial pressure is recognition logic robust to spoofing attempts, evaluated on adversarial samples — which returns the inquiry to the datasets and validation recorded in Section 3 as absent.

Saturation. The operational form of the adversarial problem is broader than perfidy and merits its own name. If recognition-for-abstention became widespread, an adversary would have an incentive not merely to feign the signal occasionally but to saturate the environment by deploying false emitters in numbers, turning each abstention into cover and each confirmed signal into noise. Deliberate saturation is the adversarial form of the false-positive problem. No signal-based scheme is immune to it; the same tactic exists, and has always existed, against the white flag itself, and the legal response has been to criminalise abuse rather than abandon the signal. HF SIGNAL 01 neither solves nor aggravates this problem. What it changes is that robustness under saturation becomes measurable rather than speculative. The adversarial model and decoy conditions specified in Section 3 are designed to measure that property, while existing design elements already bound the exposure. Dual-channel conjunction raises the cost of each false emitter above that of a stray light or bare transmission. The good-faith activation condition identifies mass false emission as abuse on the face of the specification, and the Article 41 proviso means abstention can be revoked once the supposed surrender is accompanied by hostile acts. The exposure is bounded, not eliminated. Whether the residual exposure is operationally tolerable is precisely the sort of question that belongs to validation and competent fora rather than to assertion in this paper.

One consequence of that exposure should be stated because it is a response an operational command might adopt. A detector already fitted could be disabled in software, or the relevant band could be filtered, in order to remove the tactical disadvantage. It should nevertheless be noted that deliberate disabling is a dated instruction attributable to an identifiable authority and is therefore more readily examinable than an absence of capability decided at the design stage and never recorded. Suppression of an existing recognition capability is not a return to the position that existed before the capability was introduced.

Verification of adaptive systems. Where system behaviour evolves during operation, the response interval is not a constant but a distribution, and no settled method exists for establishing that distribution merely by inspecting the system itself. This affects the available paths differently. Determining what a behavioural recogniser will do on inputs it has not seen requires reasoning about the model; determining whether a system responds to a defined signature does not, because the test emits the signature and measures the response while treating the system as opaque by design. That property is not incidental. It is what allows testing by a party without access to source code, model weights or design documentation — the normal condition of external assessment of a weapon system. The limitations of the present specification remain. A black-box test shows what the system did in the trials conducted, not what it will do in general.

Photosensitivity. The optical channel operates at 7 Hz. Current expert consensus identifies bright flashing in the 10–20 Hz range as particularly provocative for visually induced seizures, while conservative media guidance commonly restricts bright flashes above three per second. The fact that 7 Hz lies below the peak 10–20 Hz range therefore does not establish safety. Risk also depends on intensity, contrast, apparent source area, duration and duty cycle. No safety assessment of an HF SIGNAL 01 emitter has been carried out, and none is claimed here. IEC 62471-7:2023 provides a formal route for photobiological-safety assessment of visible light sources used for signalling; HFA-TN-03 accordingly treats human-exposure safety as a separate assessment and does not require deliberate viewing by human subjects. Until that question is resolved, 7 Hz remains a testable recognition value subject to a safety constraint that has not yet been established.

Equipment dependence. The active electronic modality assumes that the protected person possesses a functioning, powered emitter at the time of the encounter. The hardware is deliberately minimal, but minimal is not universal. Those most exposed to autonomous systems — displaced civilians, populations under siege and people fleeing — are precisely those least likely to carry equipment of any kind. A machine-readable standard protects only those able to "speak" it. This is a structural limitation of any active-signal approach. To mitigate it, the protocol architecture includes a passive visual modality — a white disc on a red field — requiring no

²² AP I, Art. 37(1).

power, equipment or transmission (see the accompanying Architecture Note). Although that passive modality addresses the equipment-dependence objection for those who have nothing, its machine-perception parameters — chromaticity, minimum size and viewing angle — still require specification and validation. The active electronic modality described in this paper remains subject to the equipment-dependence limit stated above. In either modality, the protocol can only be a floor for legibility, never a filter on protection, and no inference of any kind may be drawn from its absence.

Geometry of the passive modality. The passive modality is specified as a white disc on a red field. That choice is a placeholder because it was not derived from a stated perceptual or legal constraint, and this paper advances no argument in its favour. Two considerations affect its eventual determination. First is machine distinctiveness. A filled light disc on a uniform darker field is common in the ordinary visual environment — signs, logos and apertures — and the passive modality’s false-positive exposure has not been characterised. Second, independently, the pairing of white and red in a centred figure-on-field arrangement is the chromatic signature of the distinctive emblems protected under Article 38 of Additional Protocol I. This paper makes no claim of assimilation to those emblems and claims none of their protection. A protective sign intended for use by untrained persons without instruction should nevertheless be distinguishable from them at the distances and under the conditions in which both are actually read. Whether the eventual geometry should retain the present colours is a matter for a standardisation process, in which distinctiveness from Article 38 emblems belongs among the criteria alongside chromaticity, minimum size and viewing angle.

Voluntary adoption. No instrument currently in force requires a manufacturer or State to implement recognition of HF SIGNAL 01 or any other specific code. Adoption is voluntary, and this paper does not claim otherwise. Its claims end where they began. The underlying obligations exist and long pre-date the systems in question. A candidate means of giving practical effect to those obligations at machine speed now exists in documented, testable form. Whether the gap between the two is closed by this specification, by a better one, or by a standard developed within competent institutions is a question this paper can open but cannot decide.

6. Conclusion: An Invitation, Not a Product

This paper has deliberately made limited claims. It has not claimed that HF SIGNAL 01 works. No measurement exists. It has not claimed that anyone must adopt it. No instrument requires that. It has claimed that a long-standing obligation — to recognise the person who clearly expresses an intention to surrender and to do everything feasible to verify — lacks, in encounters decided at machine speed, a validated and operationally available means of implementation; that a candidate means now exists in a publicly deposited, integrity-protected form, with the active modality specified at the level of testable parameters; and that once such a candidate exists, the absence of surrender recognition is no longer only a fact about technology but also a fact about decisions.

A final clarification is owed to readers who locate the problem elsewhere — who consider autonomy in targeting itself to be the central issue, and who favour stronger human control or prohibition rather than new signals. Nothing in this paper argues for autonomous engagement. The protocol is not a licence. Recognition capability is one of several paths; it does not replace other means of meeting the obligation, and all paths are subject to the same test: whether the architecture affords, before irreversibility, the effective possibility that a legally protected change of condition is represented, recorded, and can affect action. If such systems were prohibited, or if the architecture otherwise ensured that a protected condition could be represented and acted upon before force becomes irreversible, the urgency of a machine-readable standard would diminish accordingly. This paper addresses the world in which such systems exist and are being deployed — a world in which having no means to express surrender to them protects no one and constrains no one. To readers who would prefer an argument for a standard without the candidate itself — a methodology for developing a machine-readable surrender standard rather than a completed proposal — the answer is that a standards process needs an object on which to work. Methodologies without artefacts produce committees; standards have historically begun when artefacts are submitted to processes, Annex I’s distinctive signals among them. This paper supplies such an object and, in the same act, submits it.

The natural next steps do not belong to the author. They are, first, independent technical validation by a university laboratory in machine perception or robotics, or by a standards body, against the criteria set out in Section 3, beginning with an explicit adversarial model, dataset construction and latency measurement; and second, examination in the competent fora, particularly the Group of Governmental Experts on Lethal Autonomous Weapons Systems, where machine-readable expression of surrender belongs on the agenda whether or not this particular specification survives scrutiny. The specification is offered for testing, amendment or supersession. What should not survive its publication is the claim that there was nothing to recognise.

Research materials and data status. This paper reports no experimental dataset and no performance measurements. The protocol specification and supporting technical notes identified in the notes and references provide the documentary basis for the candidate parameters and the proposed validation route. Dataset construction, detector evaluation, latency measurement and independent replication remain open tasks.

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